DS · INTELLIGENT MODELING REPORT · 2025–2026

# Lung Nodule Prognostics

Multi-task deep learning with cross-talk for pulmonary nodule risk stratification


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THE CLINICAL PROBLEM

# An indeterminate nodule is a clinical dilemma

Low-dose CT screening saves lives, but it surfaces many small lung nodules whose fate is uncertain — some stay harmless for years, others progress into aggressive cancer. Today, that call rests on a radiologist's subjective visual read.

THE CAUSE

### Subjective assessment

Risk is judged by eye from size, shape and texture — vulnerable to human error and inter-reader variability.

FAILURE MODE 1

### Overtreatment

Benign nodules sent for invasive, costly, stressful biopsies and repeated radiation exposure.

FAILURE MODE 2

### Undertreatment

Malignant tumors missed and left to grow until intervention comes too late.

WHY AN INTELLIGENT ALGORITHM

# Beyond the limits of the human eye

### Multi-task learning

Morphology, growth rate and malignancy are interdependent — not isolated variables to model in isolation.

### High-dimensional imaging

3-D CT volumes hold tiny, complex patterns the human visual system cannot extract with perfect consistency.

### Rules & linear models fall short

Progression depends on many interacting factors; rigid clinical guidelines and linear statistics cannot capture them.

### Deep learning learns hierarchy

CNNs automatically extract latent, hierarchical features straight from raw pixels and voxels.

THE CORE INSIGHT

A multi-task model can learn several biologically linked properties of a nodule at once — texture, malignancy and growth — rather than treating them as separate problems.

PREDICTION TARGETS

# Two clinical tasks — one shared model

## T1

### Malignancy risk

*Binary target*

- 0 · Benign
- 1 · Malignant

## T2

### Texture type

*3-class target*

- 0 · Solid
- 1 · Part-solid
- 2 · Ground-glass

DATA

# 508 nodule crops from LIDC-IDRI

| 1018 | DICOM | XML | 7371 |
|---|---|---|---|
| CT scans | Images | Annotations | Lesions marked as nodules |

PREPROCESSING PIPELINE

1. LIDC-IDRI Dataset
2. XML annotations
3. Texture & Malignancy & ROI
4. Crop ROI 2D

# Simplified labels — and a severe imbalance

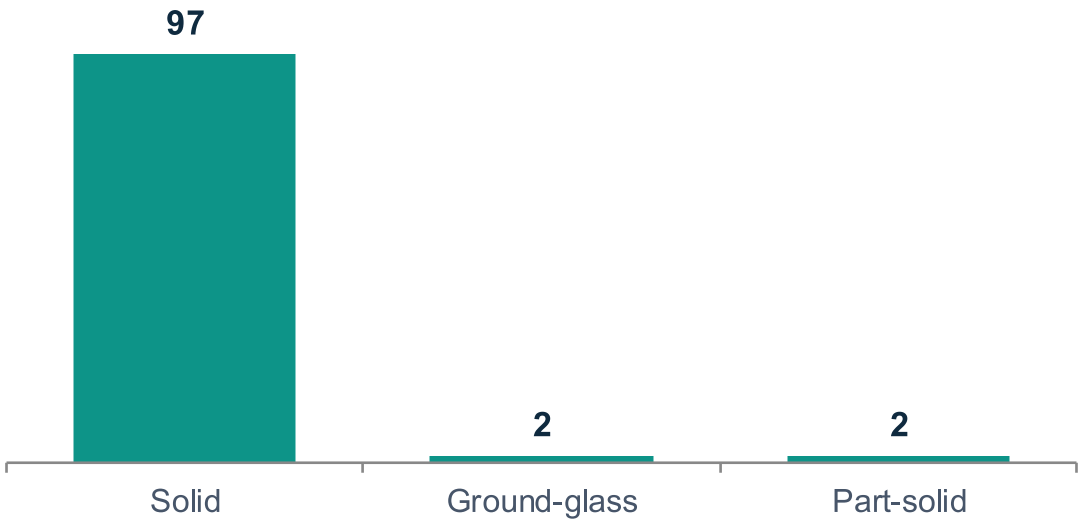
### Texture score 1–5 → 3 subtypes

**Ground-glass** scores 1–2 **Part-solid** score 3
**Solid** scores 4–5

### Malignancy → binary

**0 · Benign** **1 · Malignant**
Test split: **41 benign** / **60 malignant**

TEXTURE DISTRIBUTION · TEST SET (n = 101)




**96% solid.** With 97 of 101 test nodules solid, the minority subtypes are almost unlearnable.

EXPERIMENTAL SETUP

# How the models were trained

### Data split — 60 / 20 / 20

**306** train **101** val **101** test

### Optimizer & selection

**AdamW** · the checkpoint with the lowest validation loss is kept and scored on the held-out test set.

### Single-task baseline

Each task is trained on its own with the cross-talk switched off, so any difference comes from the sharing itself.

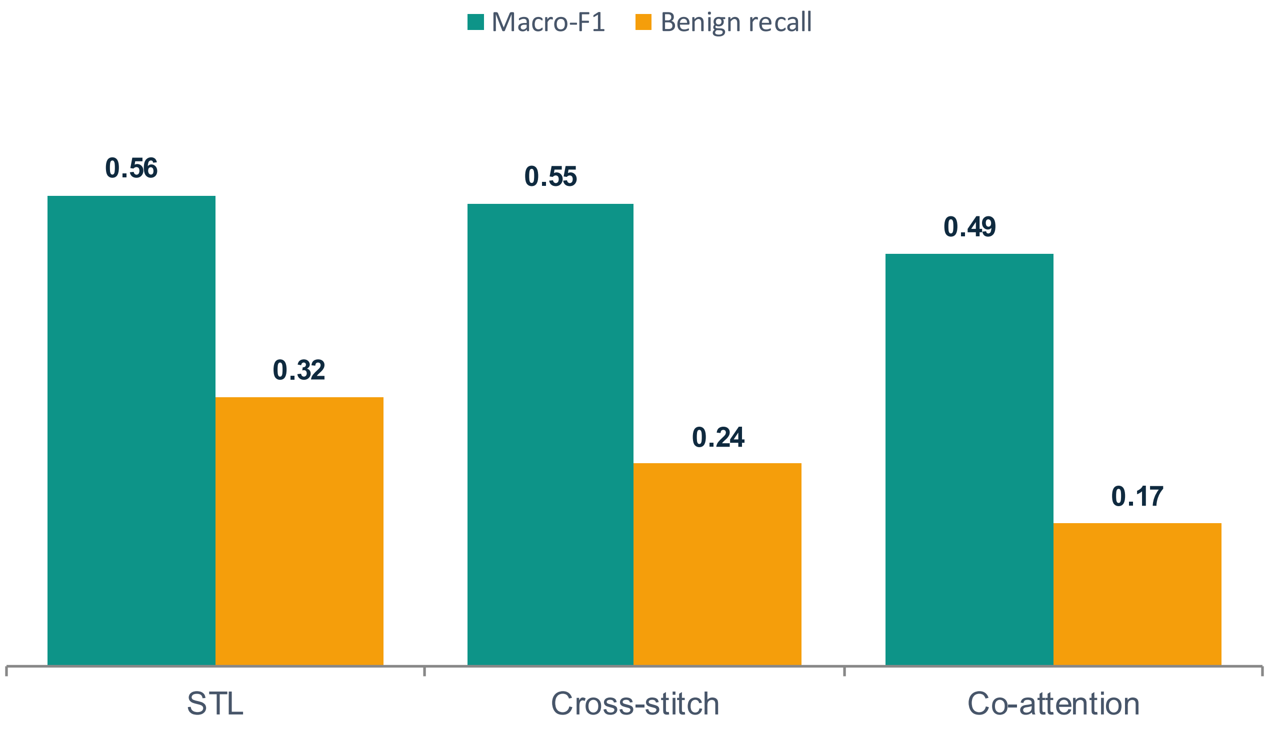
### Two sharing styles compared

**Cross-stitch** — blends both tasks' features together.
**Co-attention** — lets each task pick out the other's useful signals.

RESULTS · MALIGNANCY

# Cross-talk did not beat the baseline

MACRO-F1 vs BENIGN RECALL




| Model | Macro-F1 | Malig. F1 | Benign rec. |
|---|---|---|---|
| **STL** | 0.56 | 0.72 | 0.32 |
| **Cross-stitch** | 0.55 | 0.74 | 0.24 |
| **Co-attention** | 0.49 | 0.72 | 0.17 |

THE TRADE-OFF

Cross-stitch nudges malignant recall up to 0.90, but **erodes benign recall** — leaving macro-F1 flat. Co-attention is worse on every metric. **No clear gain from sharing.**

# All models collapse to “solid”

For texture, STL, cross-stitch and co-attention give **identical predictions** — every nodule labelled solid (macro-F1 0.33) — because 97 of 101 test samples are solid. The four held-out crops below compare ground truth (GT) with the cross-stitch model's prediction (Pred).

GT: **Solid · Benign**
Pred: **Solid · Malignant**

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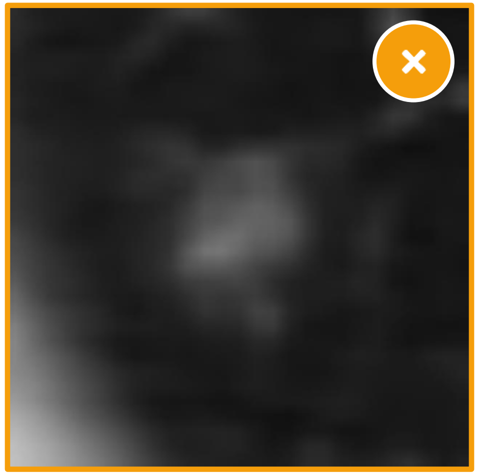

GT: **Ground-glass · Malignant**
Pred: **Solid · Benign**

**Takeaway.** With no learning gradient for the minority subtypes, the cross-talk signal carries no useful information — and can reinforce the existing majority-class bias.

THE RESULT

# The hypothesis was not supported .

### 01

### Not the architecture — the data

The bottleneck was class distribution and label quality, not the cross-talk design itself.

### 02

### Negative transfer

A degenerate texture branch (“always solid”) passed no useful gradient and reinforced the malignant-majority bias.

### 03

### Honest reporting matters

A clear negative result is a real signal for others applying multi-task learning to small, imbalanced medical datasets.

LIMITATIONS & FUTURE WORK

# What to fix, and what comes next

## Limitations

- **Extreme class imbalance —** 97 of 101 test nodules solid — minority subtypes cannot be meaningfully evaluated.
- **Modest scale —** 508 samples strain multi-branch CNNs; the richer co-attention unit likely overfits.
- **2-D only —** A single mid-slice discards the volumetric information radiologists rely on.

## Future work

- **Balance the classes —** Class-weighted loss, minority oversampling, or balanced sampling strategies.
- **Go volumetric —** 3-D convolutional inputs to capture full nodule morphology.
- **Better subtype coverage —** A larger dataset with stronger ground-glass and part-solid representation.

STRATEGIC ASSESSMENT

# SWOT analysis

## STRENGTHS

- Reproducible pipeline: DICOM → Hounsfield → normalized 128×128 crops
- Multi-task design grounded in real nodule biology
- Rigorous single-task baseline isolates the effect of sharing
- Transparent reporting of a clear negative result

## WEAKNESSES

- Severe class imbalance — 97 of 101 test nodules are solid
- Small dataset (508 crops) strains multi-branch CNNs
- 2-D single slices discard volumetric information
- Progression task dropped for lack of a data join key

## OPPORTUNITIES

- Class-balanced sampling and weighted loss
- 3-D volumetric inputs to capture full morphology
- Restore the progression task via corrected linkage
- Larger dataset with stronger subtype coverage

## THREATS

- Label noise and data quality cap achievable accuracy
- Strong existing single-task baselines are hard to beat
- Task sharing can reinforce majority-class bias
- Limited generalization beyond the LIDC-IDRI cohort

CONCLUSION

# Cross-talk is only as good as the supervision behind it.

The proposed models jointly predict nodule texture and malignancy through learnable cross-stitch and co-attention links. On 508 LIDC-IDRI crops under severe class imbalance, neither variant reliably beat the single-task baseline, the limiting factor being data quality and balance, not the architecture.